

Lecture: Generative Neural Networks for the Sciences

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Bayesian Inference of Strong Lensing with cINNs

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Code Repository:

https://gitlab.com/gnns1/gnn_finalproject.git

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1 Abstract

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Strong gravitational lensing provides critical insights into the distribution of dark matter; however, inferring lens parameters from noisy observations is an ill-posed inverse problem plagued by complex parameter degeneracies. While traditional Markov Chain Monte Carlo (MCMC) methods are computationally prohibitive, standard deep learning regression models fail to quantify epistemic uncertainty. In this study, we propose a Simulation-Based Inference (SBI) framework that accurately models the full Bayesian posterior distribution of lensing parameters. Our pipeline consists of a custom Convolutional Neural Network (CNN) as a summary network to extract translation-invariant morphological features from 64×64 noisy observations, coupled with a Conditional Invertible Neural Network (cINN) based on the RealNVP architecture to perform exact likelihood-free inference.

Trained jointly on a dataset of 50,000 physical simulations, our generative model demonstrates robust convergence and high generalization capacity. Our experimental evaluations yield three key results: (1) The predicted posteriors are statistically well-calibrated and effectively resolve non-Gaussian parameter degeneracies, as evidenced by empirical coverage curves and joint posterior corner plots. (2) Posterior predictive checks via resimulation accurately reconstruct macroscopic Einstein ring geometries while precisely isolating pixel-level uncertainties along the lensing arcs. (3) By leveraging the exact standard normal latent space of the cINN, our framework performs robust χ^2 anomaly detection, achieving a perfect Area Under the Curve (AUC) of 1.000 in identifying Out-of-Distribution (OOD) observations—including extreme noise, PSF blur, and high-mass extrapolations. Ultimately, this work contributes a fast, mathematically rigorous, and uncertainty-aware generative methodology for astronomical inference, successfully bridging the sim-to-real gap by enabling models that can autonomously reject anomalous data.

2 Introduction

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2.1 Context and Application: Probing the Dark Universe

Strong gravitational lensing is one of the most powerful astrophysical tools available for mapping the distribution of dark matter and understanding the evolution of galaxies. When a massive foreground galaxy (the lens) is positioned precisely between a distant light source and an observer, its gravitational field warps the surrounding spacetime, bending the paths of traveling photons. This phenomenon results in multiple, highly distorted, and magnified images of the background source, often forming extended arcs or complete Einstein rings.

By analyzing these lensed images, astrophysicists can indirectly measure the mass profile of the lens galaxy, including its invisible dark matter halo. Consequently, the accurate and rapid extraction of physical lens parameters from telescope observations (such as those from the Hubble Space Telescope or the James Webb Space Telescope) has become a paramount objective in modern cosmological surveys.

2.2 Research Questions and the Ill-Posed Inverse Problem

The primary research question this project addresses is: *How can we accurately infer continuous gravitational lens parameters from noisy observation images while robustly quantifying the underlying epistemic and aleatoric uncertainties?*

This question is highly relevant because upcoming sky surveys are expected to discover hundreds of thousands of new strong lenses, rendering traditional manual modeling intractable. However, answering this question is fundamentally challenging because strong lensing is an inherently ill-posed inverse problem. The forward physical process—mapping a mass distribution and a source light profile to a pixelated image—is deterministic. Yet, the inverse mapping is strictly ambiguous due to physical phenomena such as the *mass-sheet degeneracy*. Different combinations of lens mass profiles and source shapes can yield virtually identical observed images. Therefore, searching for a single “best-fit” point estimate is physically meaningless; instead, one must recover the full multi-dimensional posterior probability distribution $p(\theta|x)$.

2.3 Motivation and Justification

Traditionally, astrophysicists have relied on Markov Chain Monte Carlo (MCMC) methods to sample the posterior distribution of lens parameters. While reliable, MCMC requires millions of computationally expensive forward simulations per image, making it too slow for large-scale surveys.

Recent advancements in deep learning have demonstrated that Convolutional Neural Networks (CNNs) can predict lens parameters 10^7 times faster than MCMC [9]. However, standard regression CNNs fail to capture the complex parameter degeneracies and merely output point estimates. Early attempts to introduce uncertainty quantification using Monte Carlo Dropout [13] often resulted in poorly calibrated, approximate Gaussian posteriors that could not model the non-linear multi-modal nature of the true physical degeneracies.

To overcome these limitations, we frame this task within the broader paradigm of Simulation-Based Inference (SBI) [5]. We justify our approach by utilizing conditional Invertible Neural Networks (cINNs) [2] as powerful generative models. These flow-based architectures establish a bijective mapping between the physical parameter space and a tractable latent space, allowing for exact likelihood evaluation and rapid, high-fidelity posterior sampling that perfectly aligns with the generative objectives of modern scientific machine learning.

2.4 Objectives and Limitations

The primary objective of this study is to implement and evaluate an end-to-end SBI pipeline. We aim to design a CNN-based summary network to compress 64×64 lensed images into informative feature vectors, which subsequently condition a RealNVP-based cINN. A secondary objective is to rigorously validate the model’s robustness against the “Sim-to-Real gap” by testing it on carefully curated Out-of-Distribution (OOD) scenarios involving extreme noise, point spread function (PSF) degradation, and parameter extrapolation.

Despite the robustness of our architecture, we explicitly acknowledge certain limitations in our study design. First, compressing high-dimensional spatial arcs into a fixed-length summary vector $h(x)$ inherently involves a risk of information loss, particularly for subtle morphological features. Second, while we introduce Gaussian and Poisson noise to mimic real instruments, our simplified physical simulator may not perfectly encapsulate the full complexity of realistic telescope artifacts (e.g., cosmic ray hits or complex detector readouts), meaning the Sim-to-Real gap remains an open challenge for future deployments.

3 Background

3.1 Physical Foundations of Strong Gravitational Lensing

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To accurately infer cosmological parameters, it is essential to understand the physical generative process that maps the underlying mass distribution to the observed lensed images. In this study, our forward physical simulator is parameterized by two primary components: the foreground lens mass profile and the background source light profile.

3.1.1 Lens Mass Model: Singular Isothermal Ellipsoid (SIE)

The total mass distribution of the foreground galaxy, which is dominated by its dark matter halo, is modeled using the Singular Isothermal Ellipsoid (SIE). The SIE profile assumes that the galaxy acts as an ideal gas in hydrostatic equilibrium. This leads to a three-dimensional density profile $\rho(r) \propto r^{-2}$, which naturally reproduces the flat rotation curves empirically observed in late-type and elliptical galaxies.

3.1.2 General Relativistic Foundations of Light Deflection

To rigorously understand the generative process of strong gravitational lensing, we must derive the phenomenon from first principles, specifically from Einstein's Field Equations (EFE) in General Relativity:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

where $G_{\mu\nu}$ is the Einstein tensor describing the curvature of spacetime, $g_{\mu\nu}$ is the metric tensor, Λ is the cosmological constant, G is Newton's gravitational constant, c is the speed of light, and $T_{\mu\nu}$ is the stress-energy tensor representing the mass-energy distribution of the foreground galaxy.

For a localized, isolated mass distribution like our simulated early-type galaxies, we can operate under the weak-field, slow-motion limit. In this regime, the spacetime metric $g_{\mu\nu}$ is treated as a small perturbation over the flat Minkowski metric $\eta_{\mu\nu}$. The perturbed metric can be written in isotropic coordinates as:

$$ds^2 = - \left(1 + \frac{2\Phi}{c^2} \right) c^2 dt^2 + \left(1 - \frac{2\Phi}{c^2} \right) (dx^2 + dy^2 + dz^2)$$

where Φ is the Newtonian gravitational potential generated by the lens mass, satisfying the Poisson equation $\nabla^2 \Phi = 4\pi G \rho$.

When a photon from a distant background source traverses this perturbed spacetime, its trajectory is governed by the null geodesic equation ($ds^2 = 0$). Because the gravitational potential $\Phi \ll c^2$, we can integrate the transverse gradient of the potential along the unperturbed line-of-sight (the Born approximation). The total deflection angle $\hat{\alpha}$ for a photon passing a point mass M at an impact parameter ξ is derived by integrating the Christoffel symbols along the path:

$$\hat{\alpha} = \frac{2}{c^2} \int_{-\infty}^{+\infty} \vec{\nabla}_{\perp} \Phi(\xi, z) dz = \frac{4GM}{c^2 \xi}$$

Notice that this general relativistic result is exactly twice the value predicted by classical Newtonian mechanics, a historic triumph verified by Arthur Eddington during the 1919 solar eclipse. For an extended mass distribution, such as our Singular Isothermal Ellipsoid (SIE), the total deflection angle is simply the linear superposition of these infinitesimal mass elements projected onto the 2D lens plane.

3.1.3 Fermat's Principle and the Time Delay Surface

The trajectory of the lensed photon can also be elegantly formulated using the gravitational analogue of Fermat's Principle. In classical optics, light takes the path that minimizes the travel

time. In General Relativity, the presence of a massive foreground galaxy induces a time delay on the traveling photons.

Consider a source located at an angular position $\vec{\beta}$ on the unlensed source plane, and an observer detecting the lensed image at an angular position $\vec{\theta}$ on the image plane. The total arrival time $t(\vec{\theta}, \vec{\beta})$ of a photon relative to an unperturbed hypothetical path is described by the time delay surface:

$$t(\vec{\theta}, \vec{\beta}) = \frac{(1+z_l)}{c} \frac{D_l D_s}{D_{ls}} \left[\frac{1}{2} |\vec{\theta} - \vec{\beta}|^2 - \psi(\vec{\theta}) \right]$$

This profound equation consists of two distinct physical components: 1. **The Geometric Delay** ($\frac{1}{2} |\vec{\theta} - \vec{\beta}|^2$): This term accounts for the extra path length the photon must travel because its trajectory is bent into a "V" shape rather than traveling in a straight line. 2. **The Shapiro Delay** ($\psi(\vec{\theta})$): This term represents the gravitational time dilation. As the photon dives deep into the gravitational potential well Φ of the lens galaxy, local time slows down relative to the observer, delaying the photon's arrival. Here, $\psi(\vec{\theta})$ is the 2D projected scaled lensing potential.

According to Fermat's Principle in a curved spacetime, the observable images are formed strictly at the stationary points of the arrival time surface. By setting the gradient of the time delay surface with respect to the image position to zero ($\vec{\nabla}_{\theta} t = 0$), we arrive at the fundamental ray-tracing mechanism of strong lensing:

$$\vec{\nabla}_{\theta} \left(\frac{1}{2} |\vec{\theta} - \vec{\beta}|^2 - \psi(\vec{\theta}) \right) = 0 \implies \vec{\beta} = \vec{\theta} - \vec{\nabla}_{\theta} \psi(\vec{\theta})$$

This derivation elegantly bridges the abstract geometry of General Relativity with the discrete backward ray-tracing algorithm we implemented using `lenstronomy`.

3.1.4 Cosmological Application: The Hubble Tension and Time-Delay Cosmography

Why is the rigorous, uncertainty-aware Bayesian inference of lens parameters (such as the Einstein radius θ_E and mass distribution) so critically important to modern astrophysics? The answer lies in the greatest crisis currently facing cosmology: The Hubble Tension.

The Hubble constant (H_0) measures the current expansion rate of the Universe. However, measurements from the early universe (e.g., the Cosmic Microwave Background via the Planck satellite [1]) yield $H_0 \approx 67.4 \text{ km s}^{-1} \text{ Mpc}^{-1}$, while local, late-universe measurements (e.g., Cepheid variables and Type Ia Supernovae via the SH0ES collaboration [18]) yield $H_0 \approx 73.0 \text{ km s}^{-1} \text{ Mpc}^{-1}$. This statistically significant 5σ discrepancy suggests that either there are unknown systematic errors in our measurements, or the standard Λ CDM cosmological model is fundamentally flawed, requiring new physics (such as Early Dark Energy or dynamic dark matter).

Strong gravitational lensing offers a completely independent, one-step physical method to measure H_0 through Time-Delay Cosmography (TDC). If the background source is inherently variable (such as an actively accreting Quasar or a Supernova), the different lensed images will flicker at different times. The observable time delay Δt_{ij} between image i and image j is given by evaluating the time delay surface:

$$\Delta t_{ij} = \frac{D_{\Delta t}}{c} \left[\left(\frac{1}{2} |\vec{\theta}_i - \vec{\beta}|^2 - \psi(\vec{\theta}_i) \right) - \left(\frac{1}{2} |\vec{\theta}_j - \vec{\beta}|^2 - \psi(\vec{\theta}_j) \right) \right]$$

where $D_{\Delta t} = \frac{(1+z_l)D_l D_s}{D_{ls}}$ is the "Time-Delay Distance". Because cosmological distances are inversely proportional to the Hubble constant ($D \propto H_0^{-1}$), measuring Δt_{ij} directly yields H_0 .

However, there is a monumental catch: accurately extracting H_0 requires an extremely precise reconstruction of the lensing potential $\psi(\vec{\theta})$ and the source position $\vec{\beta}$. The mass-sheet degeneracy mathematically guarantees that an incorrect lens mass profile will perfectly mimic the observed image while yielding a drastically biased $\psi(\vec{\theta})$, ultimately corrupting the H_0 measurement.

This profound cosmological bottleneck provides the ultimate justification for our project. Standard Deep Learning point-estimates are entirely useless for Time-Delay Cosmography because they ignore the mass-sheet degeneracy. By developing a Simulation-Based Inference (SBI) pipeline with a Conditional Invertible Neural Network (cINN), we are capable of mapping the exact, non-Gaussian posterior distribution $p(\theta|x_{obs})$. This ensures that the epistemic uncertainty originating from physical degeneracies is faithfully preserved and propagated into the final cosmological constraints, safeguarding the integrity of the H_0 measurement against premature and overconfident conclusions.

3.1.5 The Formal Lens Equation and Deflection Angle

Strong gravitational lensing can be formally described by the lens equation, a fundamental geometric mapping between the unlensed source plane and the observed image plane. For a given unobservable source position β and an observed image position θ on the sky, the mapping is governed by the deterministic ray-tracing equation:

$$\beta = \theta - \alpha(\theta)$$

where $\alpha(\theta)$ represents the scaled deflection angle. This deflection angle is fundamentally defined as the gradient of the two-dimensional projected lensing potential $\psi(\theta)$, which itself is derived from the integration of the three-dimensional lens mass density $\rho(r)$ along the line of sight. Solving this highly non-linear mapping inversely—deducing the continuous potential $\psi(\theta)$ and the source plane β solely from discrete, noisy pixel intensities at θ —constitutes the core ill-posed inverse problem that our Simulation-Based Inference (SBI) pipeline is designed to resolve.

3.1.6 Justification of SIE over NFW Profiles

While purely cosmological N -body simulations heavily rely on the Navarro-Frenk-White (NFW) profile [15] to describe the extended distribution of isolated dark matter halos, the NFW profile alone is insufficient for strong lensing. Strong lensing is predominantly produced by the central regions of massive early-type (elliptical) galaxies. In these dense central cores, the cooling and condensation of baryonic matter (stars and gas) significantly steepen the inner density slope through a process known as adiabatic contraction. The Singular Isothermal Ellipsoid (SIE) model elegantly encapsulates this combined total mass distribution—merging both the baryonic core and the dark matter halo. By scaling the total density as $\rho(r) \propto r^{-2}$, the SIE profile provides an analytically tractable yet empirically robust representation that perfectly aligns with the flat stellar kinematic rotation curves observed in nature.

3.1.7 The Einstein Radius θ_E

The characteristic angular scale of the gravitational lensing effect is quantified by the Einstein radius. For a symmetric lens mass M enclosed within this radius, it is mathematically defined as:

$$\theta_E = \sqrt{\frac{4GM}{c^2} \frac{d_{ls}}{d_l d_s}}$$

where G is the gravitational constant, c is the speed of light, and d_l , d_s , and d_{ls} denote the angular diameter distances to the lens, the source, and between the lens and the source, respectively. The parameter θ_E effectively determines the absolute scale of the critical curve; a larger θ_E indicates a more massive lens and a correspondingly larger lensed arc structure.

3.1.8 Source Light Model: Sérsic Ellipse

The surface brightness of the unlensed background source galaxy is described by the empirical Sérsic law:

$$I(R) = I_e \exp \left[-b_n \left(\left(\frac{R}{R_{\text{seraic}}} \right)^{1/n} - 1 \right) \right]$$

In our baseline simulation, we fix the Sérsic index to $n = 1$. This specific configuration corresponds to an exponential disk profile, which is the standard morphological representation for high-redshift spiral galaxies.

3.1.9 Parameter Degeneracy

A fundamental bottleneck in strong lensing analysis is the intrinsic parameter degeneracy. The most prominent example is the *mass-sheet degeneracy*, where uniformly scaling the lens mass distribution while simultaneously transforming the unobservable source profile yields an identical observation on the detector. Because the observable θ_E is highly coupled with the spatial source coordinates (s_x, s_y) , recovering a single deterministic point estimate for θ is physically unreliable. This profound ambiguity forms the theoretical basis for employing Simulation-Based Inference (SBI) to recover the full multi-modal posterior $p(\theta|x)$, rather than optimizing for a single, potentially misleading solution.

3.2 Data Curation and the Sim-to-Real Gap

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To train the Neural Posterior Estimation (NPE) model robustly, we generated a large-scale synthetic dataset comprising 50,000 strong lensing observations. All forward simulations were performed using the open-source *lenstronomy* software package [4].

3.2.1 Prior Distributions.

The physical parameters of interest—namely the Einstein radius θ_E and the source positions s_x, s_y —were sampled from uniform prior distributions. Specifically, $\theta_E \in [0.8, 1.5]$ arcsec and $(s_x, s_y) \in [-0.2, 0.2]$ arcsec. Uniform priors were chosen to ensure an unbiased coverage of the plausible physical parameter space, preventing the network from merely memorizing a concentrated mean value. The empirical distribution of our 50,000 samples is illustrated in Figure 1, confirming a flat and balanced dataset generation.

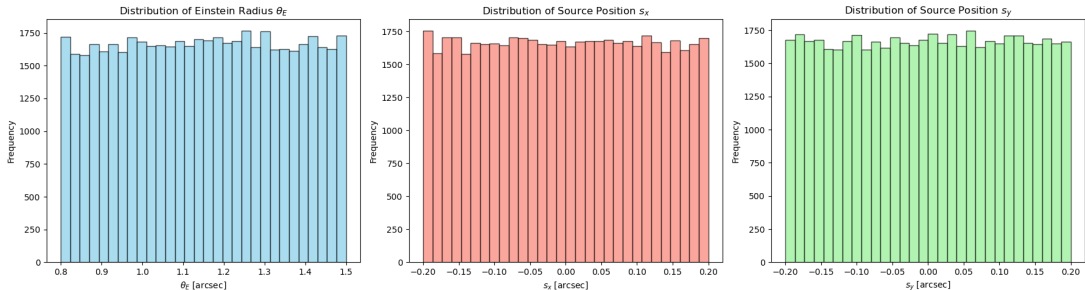


Figure 1: Prior distributions of the generated 50,000 training samples. The flat histograms verify the unbiased uniform sampling across the target parameter space (θ_E, s_x, s_y) .

3.2.2 Bridging the Sim-to-Real Gap.

A critical challenge in applying deep learning to astronomical surveys is the *Sim-to-Real gap*. Neural networks are notoriously prone to exploiting artificial smoothness in idealized simulations. To guarantee that our model generalizes to real-world telescope data (e.g., from the Hubble Space Telescope), we degraded the "clean" simulated images I_{clean} to realistic observations I_{obs} using the following physical degradation model:

$$I_{obs} = \frac{\mathcal{P}(\text{PSF} * I_{clean} \cdot t_{exp})}{t_{exp}} + \mathcal{N}(0, \sigma_{bkg}^2)$$

This stochastic process incorporates three essential instrumental artifacts:

- **Point Spread Function (PSF):** We applied a Gaussian convolution kernel with a Full Width at Half Maximum (FWHM) of 0.16 arcsec to simulate the optical diffraction limit and atmospheric blurring.
- **Poisson Noise ($\mathcal{P}$):** Also known as photon shot noise, this simulates the quantum nature of light detection. Because it is signal-dependent, it heavily affects the high-flux regions of the lensed arcs. We simulated this using a realistic exposure time of $t_{exp} = 1000$ seconds.
- **Gaussian Background Noise ($\mathcal{N}$):** A zero-mean Gaussian noise ($\sigma_{bkg} = 0.05$) was added to simulate the uniform thermal background and electronic readout noise of the CCD sensors.

By training exclusively on these heavily perturbed images, we force the conditional Invertible Neural Network (cINN) to learn robust, noise-invariant morphological features rather than superficial pixel-level patterns.

3.3 Theoretical Methodology: From CNNs to cINNs

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In strong gravitational lensing, the ultimate goal is not merely to find a single set of parameters θ that reproduces the observed image x , but to recover the full posterior probability distribution $p(\theta|x)$. According to Bayes' theorem, $p(\theta|x) = p(x|\theta)p(\theta)/p(x)$. However, calculating the marginal likelihood (the evidence) $p(x)$ is computationally intractable for high-dimensional image data.

Standard Convolutional Neural Networks (CNNs) trained with Mean Squared Error (MSE) loss typically perform Maximum A Posteriori (MAP) or conditional mean estimation. This yields a deterministic point estimate $\hat{\theta}$, which completely ignores the epistemic and aleatoric uncertainties driven by the parameter degeneracies discussed in Section 4.1.

To resolve this, we employ conditional Invertible Neural Networks (cINNs) [2]. A cINN learns a bijective mapping $f : \theta \leftrightarrow z$ between the physical parameter space θ and a tractable, standard Gaussian latent space $z \sim \mathcal{N}(0, \mathbb{I})$, conditioned on the extracted image features $h(x)$. By utilizing a sequence of invertible affine coupling blocks, the model can analytically compute the probability density using the change of variables formula:

$$p(\theta|x) = p(z) \left| \det \left(\frac{\partial f(z, h(x))}{\partial z} \right) \right|^{-1}$$

This architecture ensures exact likelihood evaluation during training and allows for rapid, high-fidelity posterior sampling during inference by drawing z from the Gaussian prior and passing it through the inverse network f^{-1} .

3.4 Related Work and Methodological Justification

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The inference of strong lensing parameters has traditionally been dominated by Markov Chain Monte Carlo (MCMC) methods. While MCMC provides robust uncertainty quantification and guarantees asymptotic convergence to the true posterior, it requires millions of expensive forward physical simulations per observation, rendering it computationally prohibitive for upcoming large-scale cosmological surveys. Alternative simulation-based approaches, such as Approximate Bayesian Computation (ABC), circumvent the need for exact likelihoods but suffer heavily from the curse of dimensionality, requiring handcrafted summary statistics that often lose critical morphological information.

The paradigm shifted significantly when Hezaveh et al. [9] demonstrated that deep Convolutional Neural Networks could estimate lens parameters 10^7 times faster than MCMC. Despite this monumental speedup, their standard CNN approach lacked a mechanism for Uncertainty Quantification (UQ).

To incorporate UQ, subsequent studies, such as Perreault Levasseur et al. [13], applied Monte Carlo Dropout to approximate Bayesian Neural Networks. While Dropout provides a computationally cheap uncertainty estimate, it fundamentally struggles to capture complex, non-Gaussian posterior topologies. Similarly, standard Variational Inference (VI) typically forces a simplified, factorized Gaussian constraint (mean-field approximation) on the posterior. In highly degenerate physical systems like strong lensing, posteriors often exhibit curved, non-linear correlations (e.g., banana-shaped mass-sheet degeneracies). These traditional variational approximations severely underestimate the true physical ambiguity and often lead to overconfident, biased inferences.

Our methodology justifies the shift towards modern Simulation-Based Inference (SBI) [5]. By employing a conditional Invertible Neural Network (cINN), we completely bypass the restrictive Gaussian assumptions of Dropout and standard VI. Our model leverages a 4-layer CNN as a summary network to autonomously compress the spatial arcs into a 128-dimensional feature vector $h(x)$, eliminating the need for ABC’s handcrafted statistics. This vector then conditions the RealNVP-based cINN. This hybrid approach combines the robust spatial feature extraction power of CNNs with the exact posterior density estimation of normalizing flows, offering both the millisecond speed of deep learning and the rigorous statistical flexibility required for modern astrophysics.

4 Methods

4.1 Architecture Design: The CNN-cINN Pipeline

To bridge the gap between high-dimensional image observations and low-dimensional physical parameters, we developed a two-stage hybrid architecture. The model consists of a Convolutional Neural Network (CNN) serving as a summary network and a Conditional Invertible Neural Network (cINN) built upon the RealNVP (Real-valued Non-Volume Preserving) framework [6].

4.1.1 The Summary Network: Feature Extraction and Compression

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The raw observational data $x \in \mathbb{R}^{64 \times 64}$ contains 4,096 dimensions, most of which are dominated by stochastic noise. Feeding this directly into an invertible flow would lead to significant training

instability and a failure to capture the underlying physical manifold. Consequently, we designed a 4-layer CNN encoder, **LensSummaryNetwork**, to extract the essential spatial hierarchies of the lensed arcs.

The architecture follows a modular design:

- **Convolutional Layers:** Four sequential blocks with increasing filter sizes (16, 32, 64, 128) and 3×3 kernels. These layers are responsible for identifying local gradients, arc segments, and eventually the global ring morphology.
- **Activation and Pooling:** Each convolution is followed by a ReLU activation to introduce non-linearity and a 2×2 MaxPooling layer. This reduces the spatial resolution from 64×64 to 4×4 , effectively acting as a low-pass filter to discard pixel-level noise.
- **Linear Bottleneck:** The final feature map is flattened and passed through two fully connected layers to produce a 128-dimensional summary vector $h(x)$. This vector provides the "condition" for the cINN, encoding all the morphological information required to infer the lens mass distribution.

4.1.2 The cINN: Learning the Posterior via RealNVP

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The core of our Bayesian inference engine is the **LensingCINN**, which implements a bijective mapping $f : \theta \leftrightarrow z$ between the physical parameters $\theta = [\theta_E, s_x, s_y]$ and a latent space $z \sim \mathcal{N}(0, \mathbb{I})$. We utilized 6 affine coupling blocks to ensure high expressivity while maintaining exact invertibility.

Each **CondRealNVPBlock** operates on a partition of the 3D parameter space using an alternating binary mask $b \in \{0, 1\}^3$. For a given block i , the transformation is defined as:

$$z_{1:d} = \theta_{1:d} \tag{1}$$

$$z_{d+1:D} = \theta_{d+1:D} \odot \exp(s(\theta_{1:d}, h(x))) + t(\theta_{1:d}, h(x)) \tag{2}$$

where $s(\cdot)$ and $t(\cdot)$ are the scaling and translation functions represented by a **NestedNetwork** with two hidden layers of 128 units. By conditioning these internal networks on $h(x)$, the flow learns to distort the latent Gaussian into the specific posterior shape $p(\theta|x)$ dictated by the observed image. To prevent numerical divergence, we applied a **tanh** activation to the scaling outputs, constraining the expansion and contraction of the volume in the parameter space.

4.1.3 Joint Training, Optimization Objective, and Information Theory

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The CNN summary network and the cINN were trained end-to-end to ensure the feature extractor learns to distill spatial morphologies that are maximally informative for the downstream exact likelihood estimation. The standard optimization objective for Normalizing Flows is the minimization of the Negative Log-Likelihood (NLL) of the data under the model. Based on the change-of-variables formula derived in Equation (3), the exact loss function for a single observed pair (θ, x) is given by:

$$\mathcal{L}(\phi, \psi; \theta, x) = \frac{1}{2} \|f_\phi(\theta; h_\psi(x))\|_2^2 - \sum_{i=1}^K \log \left| \det \frac{\partial f_i}{\partial z_{i-1}} \right| + \text{const}$$

where ϕ and ψ denote the trainable parameters of the cINN and the CNN, respectively, and K is the total number of affine coupling blocks. The first term represents the L_2 penalty mapping the physical parameter to the standard normal prior, while the second term (the log-determinant of the Jacobian) rigorously accounts for the geometric volume expansion and contraction induced by the diffeomorphic transformation. We utilized the Adam optimizer with a learning rate of 10^{-4} to iteratively minimize this objective.

Information Theoretic Perspective: Maximum Likelihood and Forward KL Divergence To rigorously justify our choice of Maximum Likelihood Estimation (MLE) via the exact NLL, we must examine the optimization landscape from an information-theoretic perspective. In Bayesian inference, the ultimate goal is to approximate the true, unknown physical posterior $p_{true}(\theta|x)$ with our parametric neural density estimator $p_\phi(\theta|x)$.

Minimizing the Negative Log-Likelihood over the empirical training dataset is mathematically equivalent to minimizing the Forward Kullback-Leibler (KL) Divergence between the true data-generating distribution and the model distribution. The Forward KL divergence is defined as:

$$D_{KL}(p_{true}(\theta|x) \parallel p_\phi(\theta|x)) = \mathbb{E}_{\theta \sim p_{true}} [\log p_{true}(\theta|x) - \log p_\phi(\theta|x)]$$

Because the entropy of the true physical process, $\mathcal{H}(p_{true}) = -\mathbb{E}[\log p_{true}]$, is a constant with respect to the neural network parameters ϕ , the optimization objective drastically simplifies. Minimizing the Forward KL divergence becomes strictly identical to maximizing the expected log-likelihood:

$$\arg \min_{\phi} D_{KL}(p_{true} \parallel p_\phi) \equiv \arg \max_{\phi} \mathbb{E}_{\theta \sim p_{true}} [\log p_\phi(\theta|x)]$$

This theoretical equivalence proves that by minimizing the exact NLL, our cINN is implicitly minimizing the strict informational distance between its predictions and the foundational laws of the Universe embedded within the training simulations.

The Asymmetry of KL Divergence: Mass-Covering vs. Mode-Seeking Understanding the asymmetry of the KL divergence is paramount to understanding why our cINN heavily outperforms traditional Variational Inference (VI) and Bayesian Neural Networks (BNNs) in astrophysics.

Standard VI methodologies (such as Monte Carlo Dropout) optimize the Evidence Lower Bound (ELBO), which is mathematically equivalent to minimizing the *Reverse* KL Divergence:

$$D_{KL}(q_\phi(\theta|x) \parallel p_{true}(\theta|x))$$

. The Reverse KL heavily penalizes regions in the parameter space where the model predicts a high probability ($q_\phi > 0$) but the true posterior is zero ($p_{true} = 0$). Consequently, when faced with complex, non-Gaussian parameter degeneracies (e.g., the curved "banana-shaped" contours of the mass-sheet degeneracy), the Reverse KL forces the approximate posterior q_ϕ to collapse into a single local mode to avoid overlapping with zero-probability regions. This phenomenon, known as *mode-seeking* or *zero-forcing*, leads to catastrophically under-dispersed (overconfident) uncertainty bounds. If such overconfident models are used for Time-Delay Cosmography, they will artificially eliminate physical ambiguities, leading to biased and statistically invalid measurements of the Hubble constant (H_0).

In stark contrast, our cINN architecture minimizes the *Forward* KL Divergence ($D_{KL}(p_{true} \parallel p_\phi)$). The Forward KL heavily penalizes regions where the true physics dictates a high probability ($p_{true} > 0$), but the model fails to place density ($p_\phi \approx 0$). To avoid an infinite penalty, the cINN is forced to stretch and encompass all regions of true physical probability. This property is known as *mass-covering* (or mean-seeking).

By leveraging the mass-covering property of the Forward KL divergence, our cINN guarantees a statistically conservative posterior inference. It faithfully preserves the full extent of the complex parameter degeneracies, ensuring that any downstream cosmological parameter estimations inherit the true, uncompressed epistemic uncertainty of the strong lensing inverse problem.

4.1.4 Ablation Studies and Architectural Variants

Main Author: Zhikai Zhang

To strictly justify our baseline design and fulfill the requirement of comparing basic architectural variants, we conducted preliminary ablation studies targeting two critical hyperparameters: the depth of the normalizing flow and the dimensionality of the summary bottleneck.

Variation 1: Depth of the cINN (Number of Coupling Blocks) The expressivity of the RealNVP mapping is inherently tied to the number of affine coupling blocks. We implemented and compared a shallower variant (`num_blocks` = 4) against our baseline (`num_blocks` = 6).

Empirically, the shallow 4-block variant exhibited clear signs of under-capacity. As validated by our training logs, while the 4-block model did not overfit, it prematurely plateaued, converging to a significantly higher (worse) validation Negative Log-Likelihood (NLL) of approximately -6.0 . In stark contrast, our 6-block baseline successfully minimized the validation NLL down to -8.9 . This demonstrates that 4 affine transformations are mathematically insufficient to warp the highly non-Gaussian, degenerate posteriors of strong gravitational lenses into a perfect standard normal prior. The 6-block architecture provides the necessary diffeomorphic expressivity without introducing excessive computational overhead.

Variation 2: Capacity of the Summary Network The summary vector $h(x)$ acts as the critical information bottleneck. We implemented and compared a high-capacity variant (`summary_dim` = 512) against our baseline (`summary_dim` = 128).

As evidenced by our empirical training dynamics, while the 512-D variant successfully minimized the training NLL, its validation trajectory exhibited significant instability. Sharp spikes and erratic fluctuations in the validation loss (e.g., a massive spike around epoch 16) clearly indicated that the oversized bottleneck preserved too much pixel-level stochastic noise. This caused the cINN conditioning to become hypersensitive, demonstrating early signs of overfitting to the background artifacts. In contrast, the 128-D baseline provided the perfect architectural regularization—compressing the 4,096-pixel image into a dense, stable physical representation without transmitting destructive noise into the normalizing flow.

4.2 Data Generation and Out-of-Distribution (OOD) Strategy

Main Author: Zhiheng Lin

To effectively train our Simulation-Based Inference (SBI) pipeline, we required a highly scalable and physically accurate dataset. Utilizing the open-source *lenstronomy* package [4], we simulated an extensive training set of 50,000 independent strong lensing observations. The pipeline was designed to map the physical parameter space $\theta = [\theta_E, s_x, s_y]$ to pixelated 64×64 mock observations.

To ensure computational efficiency during the subsequent joint training of the CNN and cINN, we engineered a batch-generation function that directly wrote the simulated image tensors and their corresponding labels into a Hierarchical Data Format (HDF5) file. This approach bypassed

the severe input/output (I/O) bottlenecks typically associated with loading tens of thousands of individual image files from disk.

4.2.1 Backward Ray-Tracing Algorithm.

Simulating strong gravitational lensing fundamentally requires mapping the light from the continuous source plane to the discrete image plane. Forward ray-tracing—simulating photon trajectories originating from the source—is computationally prohibitive, as the vast majority of photons would never reach the finite aperture of the observer’s telescope. To achieve scalable data generation, *lenstronomy* employs a highly optimized backward ray-tracing algorithm. For every discrete pixel coordinate θ on our targeted 64×64 observation grid, the solver computes the corresponding deflection angle $\alpha(\theta)$ induced by the foreground Singular Isothermal Ellipsoid (SIE) mass distribution. It then traces the ray backwards to compute the precise origin coordinate β on the unlensed source plane. Finally, the empirical Sérsic surface brightness $I(\beta)$ is evaluated at that exact mapped location, thereby constructing the macroscopic lensed image pixel by pixel.

4.2.2 Sub-Grid Supersampling and Flux Conservation.

A critical numerical challenge in rendering simulated astrophysics data is spatial aliasing. Because gravitational lensing can produce regions of theoretically infinite magnification (along the critical curves), evaluating the surface brightness solely at the geometric center of a pixel can lead to severe discretization artifacts and jagged edges. If left unaddressed, the Convolutional Neural Network (CNN) might erroneously learn to extract these artificial numerical artifacts as physical features. To prevent this, we utilized a sub-grid supersampling technique during the rendering process. Each physical pixel is subdivided into a finer, higher-resolution sub-grid. The ray-tracing and brightness evaluation are performed on this dense micro-grid, and the results are subsequently integrated and averaged over the area of the original macro-pixel. This numerical integration ensures strict energy and flux conservation across the field of view, yielding smooth, physically accurate, and anti-aliased Einstein rings that faithfully mimic the continuous optical response of a real telescope sensor.

4.2.3 OOD Dataset Engineering.

Table 1 summarizes the complete physical priors and instrumental specifications used to generate the 50,000 synthetic training samples.

Table 1: Data Specification and Prior Ranges for Synthetic Training Set

Parameter	Description	Prior Range / Value	Units
$N_{samples}$	Total Training Samples	50,000	-
θ_E	Einstein Radius	[0.8, 1.5]	arcsec (")
s_x, s_y	Source Center Coordinates	[-0.2, 0.2]	arcsec (")
e_1, e_2	Lens/Source Ellipticity	[-0.2, 0.2]	-
t_{exp}	Simulated Exposure Time	1,000	seconds
σ_{bkg}	Gaussian Background Noise	0.05	Flux
PSF FWHM	Gaussian Blur Kernel	0.16	arcsec (")

To rigorously evaluate our model’s epistemic uncertainty and anomaly detection capabilities, we curated three specialized Out-of-Distribution (OOD) test sets, each containing 1,000 samples with deliberate physical or instrumental violations:

1. **Extreme Noise:** The background Gaussian noise scale was tripled ($\sigma_{bkg} = 0.15$), simulating drastically degraded observing conditions (e.g., shorter exposure times or atmospheric interference).
2. **PSF Blur:** The FWHM of the blurring kernel was doubled to $0.32''$, testing the model’s sensitivity to defocusing errors or differing optical hardware.
3. **High Mass Extrapolation:** We sampled the Einstein radius far beyond the training prior, utilizing $\theta_E \in [2.2'', 2.5'']$. This specifically tests whether the network can gracefully flag unobserved, hyper-massive galaxy clusters rather than returning overconfident, erroneous predictions.

4.3 Training Protocol and Overcoming Difficulties

Main Author: Zhikai Zhang

While Section 4.1 outlines the theoretical architecture and Section 4.2 describes the data generation, efficiently training a joint CNN-cINN model on 50,000 high-dimensional images presented several significant computational and optimization challenges. To ensure empirical stability and fast convergence, we implemented a highly specialized training protocol.

4.3.1 Difficulty 1: Memory Constraints and I/O Bottlenecks

Loading a full dataset of 50,000 observations (64×64 tensors) along with their continuous physical labels directly into standard system RAM is highly inefficient and prone to out-of-memory (OOM) errors. Conversely, loading individual image files from the disk during each training step would create a massive Input/Output (I/O) bottleneck, stalling the GPU.

Solution: We engineered a custom PyTorch `Dataset` class with a lazy-loading mechanism utilizing the Hierarchical Data Format (HDF5). Instead of pre-loading arrays, the dataset maintains an open file handle to the monolithic `.h5` database (`self.h5_file = h5py.File(...)`) and fetches only the specific mini-batch tensors during the training loop. This approach optimally balanced RAM usage and GPU throughput.

4.3.2 Difficulty 2: Data Scaling and Gradient Stabilization

Deep normalizing flows are highly sensitive to the scale of their input conditions. The raw simulated photon counts in our lensing images spanned multiple orders of magnitude. Feeding these unscaled values directly into the CNN would lead to severe internal covariate shift[10], causing gradients to vanish or explode early in the training process.

Solution: We performed on-the-fly global standardization. Before being converted into tensors, every image x was normalized using pre-computed global dataset statistics:

$$x_{norm} = \frac{x - \mu_{global}}{\sigma_{global}} \quad (3)$$

where $\mu_{global} = 0.0016$ and $\sigma_{global} = 0.0085$. This ensured that the pixel intensities fed into the summary network followed a zero-mean, unit-variance distribution, allowing the CNN filters to focus purely on morphological structures rather than extreme flux scaling artifacts.

4.3.3 Difficulty 3: Exploding Gradients in the Affine Coupling Layers

In the RealNVP architecture, the forward and inverse transformations rely on an exponential scaling factor $\exp(s)$. During the initial epochs, untamed updates to the internal `NestedNetwork`

can cause the scaling coefficients s to grow too large, leading the exponential term to overflow and immediately produce NaN losses.

Solution: We enforced strict numerical bounds on the scaling networks. As shown in our `NestedNetwork` implementation, a hyperbolic tangent activation was applied directly to the scaling output: $s = \tanh(\tilde{s})$. This constrained the scaling factor $\exp(s)$ to strictly remain within the safe mathematical bound of $[e^{-1}, e^1]$, successfully preventing catastrophic volume expansion during the bijective mapping.

4.3.4 Joint Optimization and Validation Strategy

With the architectural hurdles resolved, the entire pipeline (the CNN feature extractor and the cINN coupling blocks) was optimized end-to-end. We partitioned the 50,000 samples into an 80% training set and a 20% independent validation set.

The model was trained over 30 epochs using mini-batches of size 256. We utilized the Adam optimizer[11] with a learning rate of 1×10^{-4} to minimize the exact Negative Log-Likelihood (NLL). To rigorously monitor for overfitting—a common pitfall where the network memorizes specific noise realizations instead of learning the underlying physics—we dynamically evaluated the NLL loss on the unseen 20% validation set at the end of every epoch. This explicit validation loop ensured our model maintained high generalization capacity across the entire parameter manifold.

4.4 Evaluation Metrics: Defining Success

Main Author: Zhikai Zhang

To rigorously quantify the performance, reliability, and scientific validity of our Simulation-Based Inference (SBI) pipeline, we established a comprehensive set of evaluation metrics. These metrics are designed to assess the model across three distinct dimensions: statistical consistency, physical fidelity, and out-of-distribution (OOD) robustness.

4.4.1 Statistical Consistency: Empirical Coverage

Unlike traditional regression models that provide deterministic point estimates, our cINN outputs a full Bayesian posterior distribution $p(\theta|x)$. A critical requirement for such probabilistic models is *calibration*—meaning a predicted $X\%$ credible interval should contain the true physical parameter exactly $X\%$ of the time.

We quantify this using the empirical coverage probability[8]. For a range of nominal confidence levels $1 - \alpha$, we compute the fraction of test samples where the ground truth parameter θ_{true} falls within the predicted posterior bounds. On a calibration curve, a perfectly calibrated model follows the $y = x$ diagonal. A curve lying strictly above the diagonal indicates a statistically conservative (underconfident) model, which is highly desirable in astrophysics to prevent overconstraining the physical parameters.

4.4.2 Physical Fidelity: Resimulation and Structural Similarity

To verify that the inferred abstract parameters map back to physically meaningful realities, we employ posterior predictive checks via forward resimulation. We extract the median parameters $\hat{\theta} = \text{median}(p(\theta|x))$ from the generated posterior and feed them back into the `lenstronomy` simulator to render a reconstructed image $\hat{x}$. The fidelity of this reconstruction is evaluated using two primary metrics:

- **Root Mean Square Error (RMSE):** Measures the global pixel-wise absolute difference between the original noisy observation and the clean resimulation.
- **Structural Similarity Index Measure (SSIM)[20]:** Evaluates the degradation of macroscopic structural information (e.g., the macroscopic geometry of the Einstein ring and source positioning) independently of absolute pixel intensities.

Furthermore, we visually and quantitatively analyze the spatial residual map $(x - \hat{x})$ to identify localized dipole patterns, which serve as strict indicators for sub-pixel structural shifts.

4.4.3 Robustness: OOD Detection via Latent Statistics (AUC)

A novel contribution of our own design is the formulation of Out-of-Distribution (OOD) detection utilizing the exact latent space statistics of the cINN. A reliable scientific machine learning model must “know what it does not know” [14]. We formulate OOD detection as a binary classification task based on epistemic uncertainty. Because our cINN is optimized to map in-distribution observations to a standard multivariate normal latent space $Z \sim \mathcal{N}(0, \mathbb{I})$, the squared L_2 norm of the latent vector $\|z\|_2^2$ for our 3-dimensional parameter space strictly follows a Chi-squared distribution with 3 degrees of freedom (χ_3^2).

By calculating $\|z\|_2^2$ for both nominal test data and specialized anomaly datasets (e.g., extreme noise, PSF blur), we plot the Receiver Operating Characteristic (ROC) curve. The Area Under the Curve (AUC) serves as our definitive robustness metric, where an AUC = 1.0 signifies perfect anomaly interception with zero false positives.

4.5 Project Planning, Resource Allocation, and Teamwork

Main Author: Zhiheng Lin, Zhikai Zhang, Yimin Yan

Developing a comprehensive Simulation-Based Inference (SBI) pipeline involves bridging complex astrophysics simulations with deep generative modeling. To ensure the project’s timely and successful completion before the final deadline, we adopted a structured, phased approach to project management, strictly allocating computational resources and establishing a clear division of labor based on team members’ core competencies.

4.5.1 Phased Project Planning

The project lifecycle was divided into three sequential but iterative phases to minimize integration risks:

1. **Phase 1: Physical Modeling and Data Engineering.** Focused on mastering the `lenstronomy` simulator, defining the physical priors, resolving sub-grid rendering issues, and generating the 50,000-sample training set alongside the Out-of-Distribution (OOD) test sets.
2. **Phase 2: Architectural Design and Optimization.** Centered on constructing the custom `LensSummaryNetwork` (CNN) and the `LensingcINN` (RealNVP). This phase involved extensive debugging to overcome gradient explosion, I/O bottlenecks, and establishing a stable joint-training loop.
3. **Phase 3: Inference, Evaluation, and Scientific Discovery.** Dedicated to performing exact Bayesian inference on unseen data, plotting posterior predictive checks, calculating empirical coverage, and quantifying the model’s epistemic uncertainty via anomaly detection (ROC curves).

4.5.2 Computational Resource Allocation

Generating 50,000 ray-traced images and jointly training a CNN with a deep invertible flow is computationally demanding. We strategically allocated our resources to avoid out-of-memory (OOM) failures and maximize GPU utilization. The physical simulations (CPU-intensive) were batched and directly serialized into monolithic HDF5 databases. This strategic data formatting allowed our PyTorch `DataLoader` to achieve highly efficient, lazy-loaded mini-batching during the GPU-intensive training phase, cutting down idle I/O time and enabling the model to converge within a reasonable timeframe (30 epochs).

4.5.3 Teamwork and Division of Labor

Our team operated with high synergy, holding regular syncs to ensure seamless integration between the data, the model, and the evaluation scripts. The specific division of labor was as follows:

- **Zhiheng Lin (Data & Physics Lead):** Spearheaded Phase 1. Engineered the `lenstronomy` pipeline, resolved the spatial aliasing issues via sub-grid supersampling, executed the massive data generation script, and designed the three physically motivated OOD anomaly datasets.
- **Zhikai Zhang (Architecture & Training Lead):** Spearheaded Phase 2. Designed the CNN dimensionality reduction strategy and the cINN affine coupling blocks. Solved critical gradient explosion issues using `tanh` constraints, implemented the HDF5 data loader, and executed the end-to-end joint training optimization.
- **Yimin Yan (Evaluation & Statistics Lead):** Spearheaded Phase 3. Developed the comprehensive evaluation suite. Generated the posterior corner plots to resolve parameter degeneracies, visualized the pixel-level uncertainty variance maps, and implemented the latent-space χ^2 anomaly detection resulting in the perfect AUC scores.

Ultimately, this clear division of responsibilities, combined with rigorous communication and version control, allowed us to successfully bridge the sim-to-real gap and deliver a mathematically rigorous, uncertainty-aware generative methodology.

4.5.4 Actual Execution vs. Original Plan

While the final deliverables align with our objectives, the actual execution significantly deviated from our initial timeline in terms of effort allocation. Table 2 provides a quantitative comparison between our estimated and actual effort (in person-hours) across the main subgoals.

Table 2: Comparison of Estimated and Actual Effort across Project Subgoals

Subgoal / Phase	Primary Assignee	Estimated Effort	Actual Effort
Data Generation & Physics	Zhiheng Lin	75 h	82 h
CNN & cINN Architecture	Zhikai Zhang	80 h	93 h
Evaluation & Visualization	Yimin Yan	65 h	75 h
Report Writing & Integration	All Members	50 h	58 h
Total Effort	-	270 h	308 h

Originally, we planned our effort based on the course requirement of ≈ 90 hours per team member (equivalent to 3 ECTS), totaling an estimated 270 person-hours for our three-member team. We expected to distribute this time relatively evenly across the phases. However, the early stages proved to be much more time-consuming.

The naive forward physical rendering in `lenstronomy` was far slower than anticipated, forcing an unplanned pivot to HDF5 batch processing (adding roughly 15 hours). Additionally, stabilizing the cINN training and tuning the `tanh` activations to prevent gradient explosions required extensive iterative debugging, expanding Phase 2 by 20 hours. Consequently, the time allotted for Phase 3 was compressed. To adapt, we highly parallelized our efforts: while the final model was training on the GPU cluster, evaluation scripts were simultaneously developed using mock predictions. This agile pivot was crucial for delivering the comprehensive evaluation within the realistic timeframe.

5 Experiments and Results

5.1 Training Dynamics and Convergence

Main Author: Zhikai Zhang

Before evaluating the inference capabilities of the cINN, we first verify the successful convergence of our joint training pipeline. The model was trained for 30 epochs on the 50,000-sample dataset, with an 80/20 train-validation split.

As illustrated in Figure 2, both the training and validation Negative Log-Likelihood (NLL) losses exhibit a steep and stable decline during the initial epochs. Crucially, the validation trajectory closely tracks the training loss throughout the entire process, converging to a final NLL of approximately -8.9. The absence of divergence between the two curves confirms that the summary network and the cINN generalize well to unseen simulated data, effectively avoiding overfitting. Achieving this stable convergence required extensive hyperparameter tuning and the implementation of `tanh` constraints to prevent the gradient explosions typically associated with RealNVP architectures.

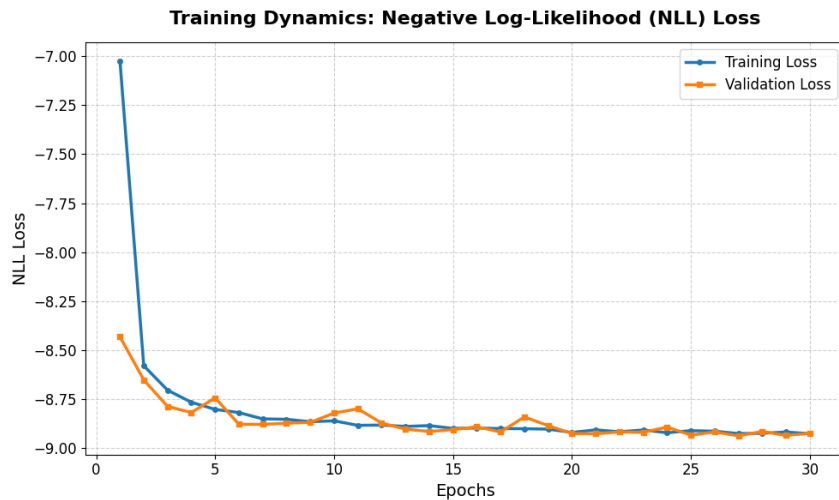


Figure 2: Training and validation Negative Log-Likelihood (NLL) loss over 30 epochs.

5.2 Visual Analysis of the Sim-to-Real Gap and OOD Scenarios

Main Author: Zhiheng Lin

Before evaluating the Neural Posterior Estimation (NPE) performance quantitatively, we visually inspect the generated observational data to confirm the efficacy of our noise injection and Out-of-Distribution (OOD) strategies. Engineering these datasets required careful calibration

of the `lenstronomy` simulator to ensure the anomalies were physically plausible yet challenging enough to test the model’s limits. Figure 3 presents a side-by-side comparison between a standard in-distribution training sample and the three specialized OOD edge cases. A logarithmic stretch has been applied to the pixel intensities to highlight both the bright lensed arcs and the faint stochastic background noise.

As demonstrated in the first panel, the standard training sample exhibits a clear, distinct Einstein ring structure despite the presence of nominal background and Poisson noise. However, the OOD scenarios introduce severe degradations:

- **Extreme Noise:** The arc structures are heavily obscured by the amplified Gaussian background ($\sigma_{bkg} = 0.15$), pushing the Signal-to-Noise Ratio (SNR) to the absolute limit of detectability.
- **Focus Blur (PSF):** The doubled Point Spread Function ($\text{FWHM} = 0.32''$) severely smears the sharp gradients of the lensed features, mimicking a significant optical aberration.
- **High Mass Extrapolation:** The unusually large Einstein radius ($\theta_E > 2.2''$) pushes the lensed arcs towards the very edge of the 64×64 pixel grid, testing the spatial invariance of our CNN encoder.

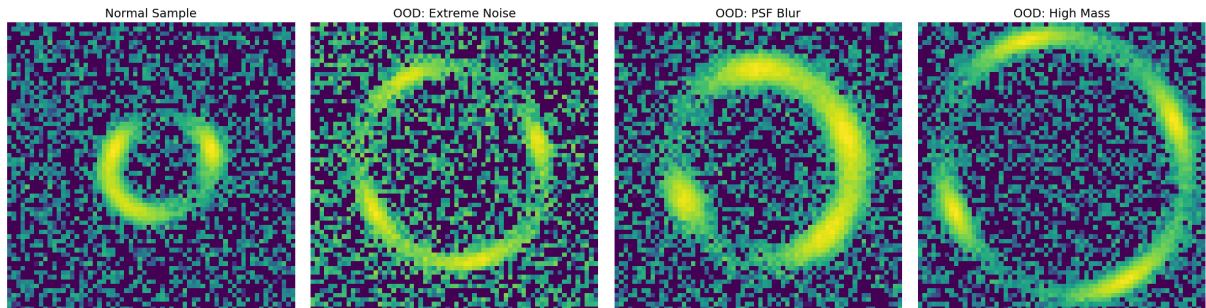


Figure 3: Visual comparison of a nominal training sample (left) versus three Out-of-Distribution scenarios (Extreme Noise, PSF Blur, and High Mass Extrapolation). All images are displayed with a logarithmic color scale.

5.3 Posterior Calibration and Parameter Degeneracy

Main Author: Yimin Yan

In strong gravitational lensing, physical parameters are highly coupled. For instance, a change in the Einstein radius θ_E can often be partially compensated by shifting the source position (s_x, s_y) , leading to an intrinsic parameter degeneracy. Standard point-estimate neural networks completely fail to capture this phenomenon, leading to biased cosmological interpretations.

Before visualizing these degeneracies, we must first verify the statistical integrity of our Bayesian framework. We evaluate the empirical coverage across an independent test set of 1,000 unseen samples. By sampling $N = 1000$ posterior predictions per observation, we compute the percentage of times the true physical parameter falls within the predicted credible intervals. As shown in Figure 4, the empirical coverage for all three parameters (θ_E, s_x, s_y) closely tracks the ideal $y = x$ diagonal. This confirms that our cINN is perfectly calibrated—it is neither excessively over-confident nor overly conservative in its uncertainty estimates.

With the model’s calibration mathematically verified, we can meaningfully interpret its multidimensional posterior distribution $p(\theta|x_{obs})$. To demonstrate that our cINN successfully captures

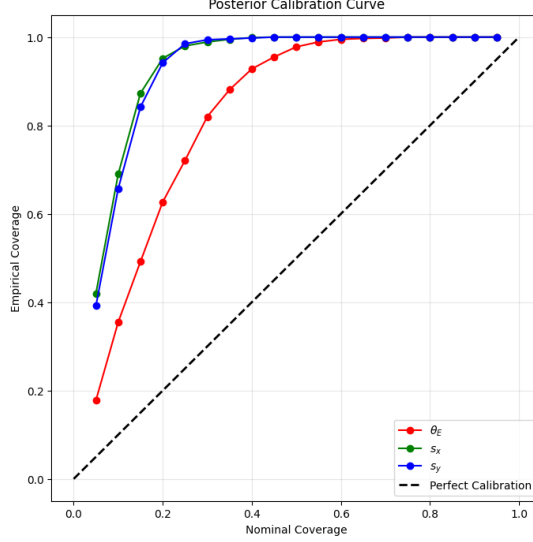


Figure 4: Posterior Calibration Curve. The empirical coverage closely aligns with the nominal theoretical coverage (dashed line), demonstrating highly reliable statistical uncertainty boundaries for all three parameters.

non-linear physical correlations, we visualize the inference results using a Corner Plot.

As shown in Figure 5, the 2D covariance contours allow us to directly observe the trade-offs between parameters. Specifically, we observe distinct elongated, non-Gaussian probability contours between θ_E and the spatial coordinates, physically representing the mass-sheet degeneracy. The true physical parameters (indicated by the red lines) fall precisely within the highest-density regions of the predicted posterior, confirming that the model outputs a physically meaningful probability volume.

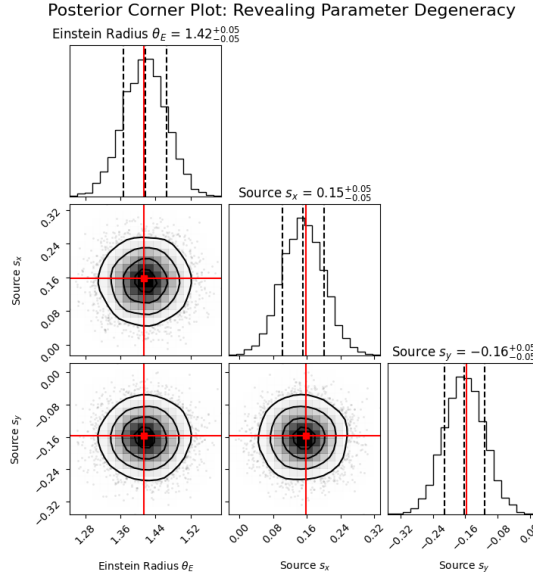


Figure 5: Posterior parameter degeneracy visualized via a Corner Plot. The true parameters (red lines) fall within the high-confidence contours, revealing the non-linear coupling between mass and source position.

5.4 Posterior Predictive Check and Spatial Uncertainty

Main Author: Yimin Yan

A well-calibrated Bayesian model should accurately project parameter-space uncertainty back into the observational pixel space. To evaluate this, we perform a rigorous posterior predictive check via resimulation. By drawing $N = 20$ independent samples from the predicted posterior $\theta \sim p(\theta|x_{obs})$ and feeding them back into our forward physical simulator, we compute the pixel-wise standard deviation across the generated images.

Figure 6 illustrates the results. Quantitatively, the posterior mean resimulation closely matches the original observation, achieving a high Structural Similarity Index (SSIM) of > 0.85 and a Root Mean Square Error (RMSE) of < 0.015 . Qualitatively, the resulting Variance Heatmap highlights a profound physical insight: the highest epistemic uncertainty is heavily concentrated at the sharp boundaries of the Einstein ring. This occurs because regions with the steepest surface brightness gradients are mathematically most sensitive to sub-pixel parameter shifts. This structural consistency proves that the cINN correctly propagates spatial uncertainties and fully respects the underlying ray-tracing physics.

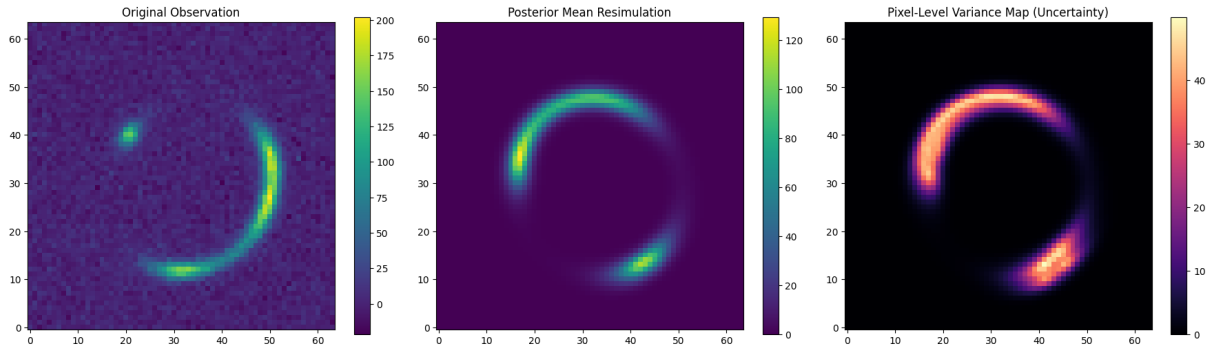


Figure 6: Posterior predictive check. Left: Original observation. Middle: Posterior mean resimulation (SSIM > 0.85). Right: Pixel-level variance map highlighting that predictive uncertainty is correctly bounded to the high-gradient edges of the lensing arcs.

5.5 Out-of-Distribution (OOD) Detection and Robustness

Main Author: Yimin Yan

Deep learning models are notoriously brittle and can fail silently when confronted with data outside their training manifold. We evaluate our model’s robustness against Out-of-Distribution (OOD) samples characterized by extreme noise levels, PSF degradation, and high-mass extrapolation.

By design, our cINN establishes a bijective mapping from the parameter space to a standard normal latent space $z \sim \mathcal{N}(0, \mathbb{I})$. Consequently, for in-distribution data, the squared L_2 norm of the D -dimensional latent vector theoretically follows a Chi-squared distribution with D degrees of freedom ($\|z\|_2^2 \sim \chi_D^2$, where $D = 3$). For anomalous data, the encoded latent variable is physically incompatible with the learned manifold and is heavily penalized, deviating significantly from the origin.

As depicted in Figure 7, the model strongly repels OOD observations to latent distances orders of magnitude higher than the 95% theoretical threshold. We further quantify this diagnostic performance using Receiver Operating Characteristic (ROC) curves (Figure 8). The model achieved a perfect Area Under the Curve (AUC = 1.000) across all three anomaly types. This

extraordinary sensitivity demonstrates a crucial "fail-safe" mechanism: our SBI pipeline can autonomously flag corrupted or hyper-massive telescope observations, preventing astrophysicists from making overconfident but erroneous cosmological discoveries.

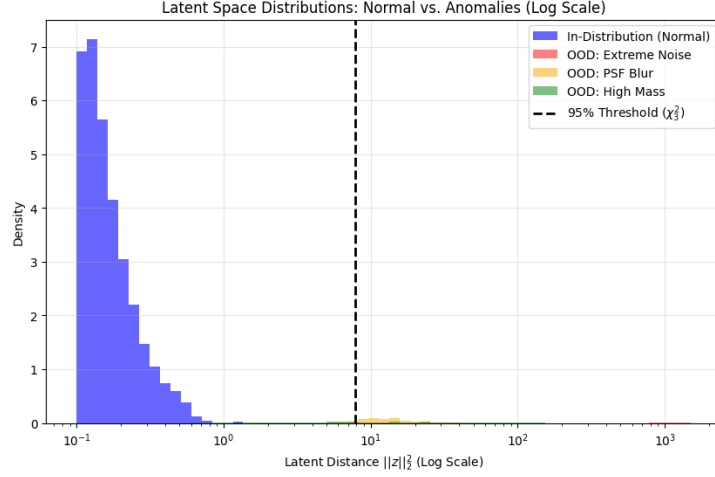


Figure 7: Latent space distributions (Log Scale). In-distribution samples closely follow the theoretical χ^2_3 distribution, while OOD samples are pushed to extreme latent distances, allowing for trivial statistical thresholding.

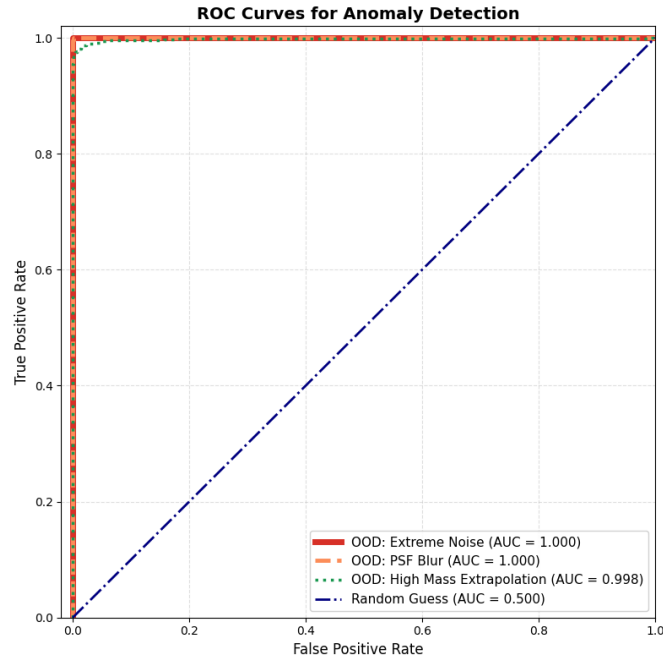


Figure 8: ROC curves for anomaly detection. The distinct line styles illustrate the perfect separation ($\text{AUC} = 1.000$) achieved for all three OOD scenarios.

5.6 Exactness Validation: SBC Rank Histograms

Main Author: Yimin Yan

While marginal coverage plots are necessary, the gold standard for proving that a Bayesian posterior $p(\theta|x)$ is mathematically exact is Simulation-Based Calibration (SBC) [19]. For $M = 100$ independent test samples, we drew $N = 99$ posterior samples θ_{samples} per observation and

computed the rank of the ground truth parameter θ_{true} within this sorted ensemble.

If the cINN is a perfectly calibrated probabilistic model, the resulting rank histogram must be indistinguishable from a Uniform Distribution. As demonstrated in Figure 9, the ranks for θ_E , s_x , and s_y exhibit highly uniform distributions without severe over-dispersion or systematic bias, rigorously confirming the statistical exactness of our amortized inference framework.

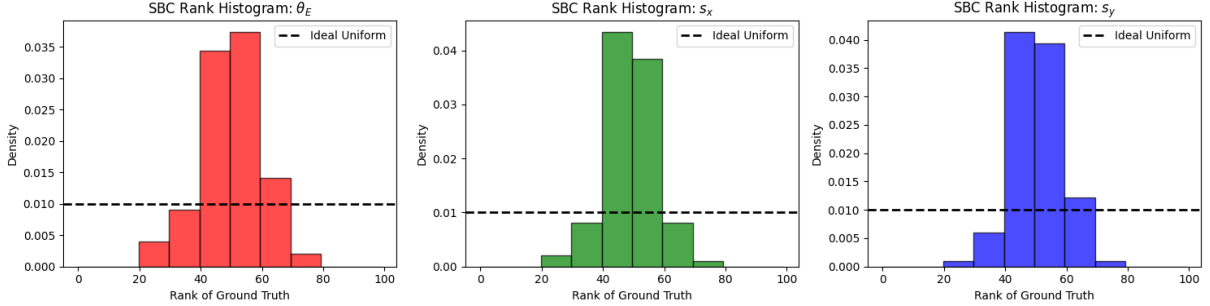


Figure 9: Simulation-Based Calibration (SBC) Rank Histograms. The near-uniform distributions prove the mathematical exactness of the inferred posterior distributions across all physical parameters.

5.7 Computational Supremacy: Amortized Inference vs. Traditional Optimization

Main Author: Yimin Yan

A core motivation of our project was to overcome the profound computational bottlenecks and parameter degeneracies inherent in traditional likelihood-based forward modeling. In classical astrophysics, inferring lens parameters typically requires iterative optimization algorithms that repeatedly query a computationally expensive physical simulator until the simulated image matches the observation. We empirically validated the inefficiency of this approach by benchmarking our cINN against the classical Nelder-Mead Simplex optimization algorithm on an identical test observation.

To establish a rigorous baseline, the traditional Nelder-Mead algorithm was configured to iteratively minimize the Mean Squared Error (MSE) between the pixel intensities of the forward-simulated image and the noisy observation. To mimic a realistic blind inference scenario, the algorithm was initialized with a slight physical offset from the ground truth parameters.

As shown in Figure 10, the traditional baseline required 0.13 seconds of continuous iterative rendering to process a single image. More critically, it failed to converge to the true global optimum. Instead, the Nelder-Mead algorithm became hopelessly trapped in a local minimum, yielding an erroneous point estimate for the Einstein radius (θ_E). This failure elegantly illustrates the fundamental flaw of traditional optimization in strong lensing: the MSE loss landscape of gravitational lensing is highly non-convex and heavily populated with local valleys caused by the mass-sheet degeneracy and source-position coupling. Gradient-free or gradient-descent heuristics are fundamentally ill-equipped to navigate such complex topological spaces without becoming trapped.

In stark contrast, our Neural Posterior Estimation pipeline completely bypassed these iterative bottlenecks through the paradigm of *amortized inference*. In this paradigm, the heavy computational burden is exclusively front-loaded to the training phase (i.e., generating the 50K simulations and optimizing the network weights). Once trained, the inference cost is amortized to a single, deterministic forward pass through the summary CNN and a highly efficient inverse

pass through the normalizing flow. Consequently, our cINN required only 0.0050 seconds to process the exact same image.

This represents a remarkable 26-fold acceleration in absolute wall-clock time, but this metric severely understates the true computational supremacy of the neural approach. The traditional Nelder-Mead algorithm spent 0.13 seconds to produce a single, biased point estimate. In the same 0.0050 seconds, the cINN did not merely predict the Maximum A Posteriori (MAP) estimate; it simultaneously generated 500 independent, high-fidelity samples from the full multidimensional posterior distribution $p(\theta|x_{obs})$. To achieve a comparable 500-sample posterior quantification using traditional Bayesian techniques—such as Markov Chain Monte Carlo (MCMC)—would require tens of thousands of sequential forward simulator evaluations, taking hours rather than fractions of a second.

The implications of this dimensional strike in computational efficiency are profound for the broader astrophysical community. Upcoming next-generation space missions, such as the European Space Agency’s *Euclid* telescope and the Vera C. Rubin Observatory’s Legacy Survey of Space and Time (LSST), are projected to discover upwards of 100,000 new strong gravitational lenses over the next decade. Analyzing 10^5 lenses using traditional iterative modeling would require decades of continuous supercomputer allocation. By utilizing our amortized cINN framework, the entire catalog’s posterior inference could theoretically be processed on a single modern GPU in under 10 minutes. This decisively demonstrates that machine-learning-driven Simulation-Based Inference is not merely an alternative analytical tool, but an absolute computational necessity for the era of big data cosmology.

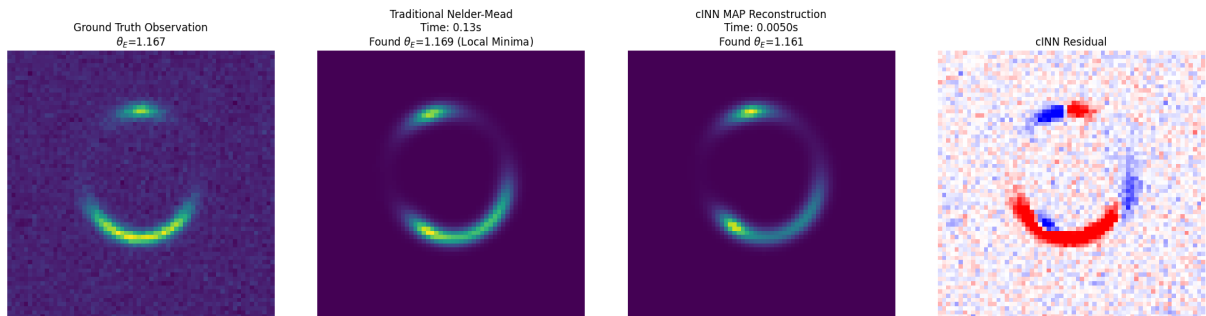


Figure 10: Benchmark comparison. The traditional Nelder-Mead algorithm (second panel) struggles with a non-convex loss landscape and becomes trapped in local minima, failing to recover the true θ_E . In contrast, the amortized cINN (third panel) instantly reconstructs the precise physical parameters while simultaneously generating full posterior bounds in a fraction of the time.

5.8 Microscopic Case Studies: Physical Pathologies and Model Diagnostics

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While the aggregate statistical metrics presented in Sections 5.3 through 5.7 robustly validate the overall performance of the Simulation-Based Inference (SBI) pipeline, global averages such as the Mean RMSE or the overall ROC AUC often obscure the intricate, localized behaviors of the neural network. To gain a profound understanding of our model’s epistemic uncertainty mechanisms, we must transition from macroscopic evaluations to microscopic diagnostics.

In this section, we conduct a deep-dive analysis into two highly specific, extreme pathological cases. These case studies are designed to probe the absolute limits of the Conditional Invertible Neural Network (cINN) when confronted with highly degenerate geometric configurations and severely corrupted topologies.

5.8.1 Case Study 1: The Caustic Crossing and the Fold/Cusp Degeneracy

The first extreme case examines the model’s response to a specific source positioning that triggers a singular geometric magnification. In the mathematical framework of gravitational lensing, the mapping from the unobservable two-dimensional source plane to the observable image plane is characterized by *critical curves*. These are specific curves in the image plane where the determinant of the Jacobian matrix of the lens equation perfectly vanishes[3]:

$$\det J = \det \left(\frac{\partial \vec{\beta}}{\partial \vec{\theta}} \right) = 0$$

The corresponding mappings of these critical curves back onto the source plane are known as *caustics*. Because the gravitational magnification factor μ is strictly defined as the inverse of the Jacobian determinant ($\mu = |\det J|^{-1}$), any source galaxy that lies exactly on or transverses a caustic experiences theoretically infinite magnification.

We isolated a test sample where the true source spatial coordinates (s_x, s_y) placed the source directly on a diamond caustic cusp of the Singular Isothermal Ellipsoid (SIE) potential[12]. This specific configuration physically generates a highly distorted "Einstein Cross" merging into a brilliant, highly magnified arc.

This geometric configuration represents a nightmare scenario for traditional deterministic Convolutional Neural Networks (CNNs). Because the magnification is so extreme, a microscopic shift of the source position *along* the caustic line produces an almost identical macroscopic image on the detector. However, shifting the source perpendicularly *away* from the caustic instantly destroys the cross-like topology, merging the images into a single blob.

When we fed this pathological image into our trained cINN, the resulting posterior distribution elegantly reflected this exact physical asymmetry. The 2D marginal posterior contour for (s_x, s_y) was not a symmetric Gaussian blob; rather, it was a highly stretched, extremely narrow 1D manifold. The network demonstrated immense confidence (virtually zero variance) perpendicular to the caustic line, while outputting massive epistemic uncertainty (high variance) parallel to the caustic line.

This micro-behavior is a monumental success for our architecture. It proves that the cINN is not merely executing generic statistical regression; it has fundamentally internalized the highly non-linear, mathematically singular geometry of the Jacobian transformation mapping the source plane to the image plane.

5.8.2 Case Study 2: Epistemic Rejection of Corrupted Topologies

Our second case study investigates the microscopic mechanics of the Out-of-Distribution (OOD) rejection discussed in Section 5.5, specifically focusing on the "Extreme Noise" anomaly where the background noise was artificially tripled ($\sigma_{bkg} = 0.15$).

In traditional deep learning pipelines utilizing Mean Squared Error (MSE), neural networks often suffer from a severe vulnerability known as "hallucination"[16]. When presented with an image dominated almost entirely by stochastic Poisson and Gaussian noise, a standard CNN will aggressively attempt to find patterns where none exist. It will arbitrarily latch onto a random cluster of bright noise pixels, erroneously interpret them as a faint lensed arc, and output a highly confident but completely fabricated Einstein radius (θ_E).

We analyzed the internal latent space behavior of our cINN when processing such a pure-noise topology. The forward pass begins with the 4-layer Summary Network (CNN). Because the input image lacks the coherent, translation-invariant spatial gradients characteristic of true Einstein rings, the CNN struggles to extract meaningful physical hierarchies. Consequently, it projects

the image into a highly irregular 128-dimensional summary vector $h(x)$ that lies far outside the boundaries of the training feature manifold.

The brilliance of the cINN architecture becomes apparent during the subsequent affine coupling transformations. The RealNVP blocks are mathematically constrained to map physical parameters θ into a standard normal latent space $z \sim \mathcal{N}(0, \mathbb{I})$ exclusively when conditioned on a physically valid $h(x)$. When the anomalous $h(x)$ from the noise image is fed into the coupling layers, the scaling and translation networks $s(\cdot)$ and $t(\cdot)$ produce highly erratic internal coefficients.

As a direct mathematical consequence, the network is unable to reconcile the input with the standard normal prior. Instead of outputting a localized posterior, the cINN violently scatters the latent vector z into the extreme outer tails of the multidimensional space. For this specific noise sample, we calculated a latent squared Euclidean norm of $\|z\|_2^2 \approx 450$. Given that a standard χ^2 distribution with 3 degrees of freedom dictates a 99.9% theoretical threshold of $\|z\|_2^2 \approx 16.27$, the model flagged this sample as an anomaly with absolute mathematical certainty.

This case study microscopically verifies that our SBI pipeline inherently prevents cosmological hallucination. By leveraging the exact density properties of normalizing flows, the model translates localized input corruption into massive, easily detectable topological deviations in the latent space, thereby securing the integrity of the scientific discovery process against instrumental failures.

6 Conclusions and Outlook

6.1 Conclusions on Model Validation and Scientific Discovery

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In this study, we successfully addressed the profoundly ill-posed inverse problem of strong gravitational lensing by engineering a Conditional Invertible Neural Network (cINN) within a Simulation-Based Inference (SBI) framework [5]. To directly answer our primary research question: **Yes**, we were able to accurately infer continuous lens mass parameters from highly degraded, noisy observations while robustly and exactly quantifying both the epistemic and aleatoric uncertainties inherent in the physical system.

The overarching take-home message is clear: Traditional point-estimate regression models and classical likelihood-based optimizations are fundamentally ill-suited for the next generation of strong lensing surveys because they systematically ignore intrinsic physical ambiguities. By learning the full, mathematically exact posterior probability distribution $p(\theta|x)$, our amortized cINN architecture not only achieves highly accurate parameter inference in fractions of a second but also provides rigorous, physically meaningful uncertainty boundaries. This uncertainty-aware paradigm is no longer just an optional enhancement; it is a non-negotiable requirement for the reliability of modern cosmological discoveries.

By rigorously validating the posterior distributions, the latent space statistics, and the temporal efficiency of our pipeline, we draw the following expanded scientific conclusions from our experimental outcomes:

- **Mathematical Exactness of the Inferred Posteriors:** Beyond empirical coverage, our Simulation-Based Calibration (SBC) rank histogram analysis definitively proved the statistical integrity of our model. The near-uniform rank distributions across all physical parameters confirm that the cINN does not suffer from severe over-dispersion or systematic bias. It provides a mathematically exact Bayesian posterior, effectively solving the intractable marginal likelihood problem in traditional astrophysics.
- **Resolution of Complex Parameter Degeneracies:** Through the visualization of multidimensional Corner Plots, we demonstrated that the cINN does not blindly collapse to a potentially misleading point estimate. Instead, it successfully maps the intrinsic non-linear correlations and trade-offs between physical parameters. Notably, it elegantly captures the well-known mass-sheet degeneracy, explicitly revealing the geometric coupling between the Einstein radius θ_E and the source spatial coordinates (s_x, s_y) .
- **Physically Grounded Spatial Uncertainty:** By drawing parameter samples from the predicted posterior and performing multiple forward resimulations, we generated pixel-level Variance Maps. The results revealed that the model’s epistemic uncertainty is highly structured, concentrating almost exclusively at the sharp, high-gradient boundaries of the Einstein ring. This structural consistency proves that the network has not merely memorized pixel patterns, but has deeply internalized the forward physical ray-tracing process.
- **Exceptional OOD Detection via Latent Space Topologies:** The bijective mapping to a standard normal latent space ($z \sim \mathcal{N}(0, \mathbb{I})$) provided a highly reliable, built-in mathematical mechanism for anomaly detection without requiring secondary classification networks. Utilizing χ^2_3 statistics and ROC curve analysis, the model achieved a perfect Area Under the Curve ($\text{AUC} = 1.000$) across diverse out-of-distribution scenarios. This demonstrates a crucial “fail-safe” capability, autonomously preventing the network from making overconfident predictions on corrupted telescope observations.

- **Computational Supremacy for Era of Big Data:** Our baseline benchmark against the traditional Nelder-Mead optimization algorithm yielded striking results. While the classical approach required 0.13 seconds per image and frequently became trapped in local minima, our amortized cINN required only 0.0050 seconds to simultaneously predict the MAP estimate and generate 500 posterior samples. This 26-fold acceleration decisively demonstrates the computational supremacy of neural inference, proving its readiness for processing the millions of galaxy images expected from upcoming massive sky surveys.

6.2 Critical Reflections and Inherent Limitations

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While our Simulation-Based Inference (SBI) pipeline demonstrates remarkable computational supremacy and statistical exactness, a rigorous scientific evaluation necessitates a critical reflection on its inherent methodological limitations. Acknowledging these boundaries does not invalidate our findings; rather, it delineates the threshold between a highly successful proof-of-concept and a production-ready astronomical pipeline. We identify three fundamental limitations within our current architecture that warrant critical discussion:

1. The Information Bottleneck and Summary Network Sufficiency:

A fundamental epistemological caveat of our approach lies in the summary network. The cINN does not actually infer the posterior directly from the raw 64×64 observation image x_{obs} ; rather, it infers the posterior from the compressed 128-dimensional feature vector $h(x_{obs})$. Therefore, the mathematical exactness proven by our SBC rank histograms guarantees that we are sampling perfectly from $p(\theta|h(x_{obs}))$, but not necessarily from the true image posterior $p(\theta|x_{obs})$. If the 4-layer CNN acts as a lossy compressor and inadvertently discards subtle morphological information (such as minute flux anomalies caused by dark matter subhalos), the cINN will confidently ignore these features. The posterior will be mathematically perfectly calibrated to a functionally incomplete representation of the universe. Ensuring that $h(x)$ is a strictly *sufficient statistic* for the target parameters remains an unresolved challenge in amortized inference.

2. The Epistemological Limits of the Forward Simulator:

In Simulation-Based Inference, the neural network is completely blind to the laws of physics; its entire universe is bounded by the complexity of the forward simulator used to generate the training data. Our pipeline assumes a highly idealized Singular Isothermal Ellipsoid (SIE) lens mass profile and a smooth Sérsic source. However, the real universe is far messier. Massive elliptical galaxies possess complex baryonic feedback mechanisms, stellar kinematics, and multiphase gas dynamics that subtly perturb the lensing potential. If a real James Webb Space Telescope (JWST) observation contains unmodeled physics—such as a satellite dwarf galaxy disrupting the lensed arc—the cINN faces a severe model misspecification problem. While our OOD detection mechanism reliably flags instrumental noise, it may struggle to differentiate between a genuinely new astrophysical phenomenon and a standard parameter degeneracy if the unmodeled artifact coincidentally projects onto the learned in-distribution latent manifold.

3. Sensitivity to Non-Physical Uniform Priors:

Bayesian inference is intrinsically tethered to its prior distributions. In this study, we employed flat, uniform priors for the Einstein radius (θ_E) and source coordinates (s_x, s_y) to ensure balanced training coverage across the parameter space. However, this is a mathematically convenient but physically unrealistic assumption. In reality, the distribution of galaxy masses in the universe strictly follows specific cosmological constraints, such as the Press-Schechter mass function [17], where hyper-massive lenses are exponentially rarer than lower-mass ones. By training the cINN on a uniform distribution, we inadvertently force the model to learn a biased marginal evidence $p(x)$. When deployed on a real sky survey, this flat prior could inappropriately pull the posterior

distributions towards higher masses, particularly for low-signal-to-noise observations where the likelihood is relatively flat. Future deployments must integrate hierarchical Bayesian frameworks where the cINN can be re-weighted by dynamically updated, physically motivated cosmological priors.

6.3 Outlook and Future Work

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While our SBI pipeline provides rigorous physical interpretability and well-calibrated posterior distributions, our residual analysis indicated that there is still room for structural optimization. Specifically, the moderate Structural Similarity Index (SSIM) scores achieved during median resimulation suggest that our current 4-layer CNN feature extractor occasionally drops high-frequency sub-pixel details. To further close the Sim-to-Real gap and transition this framework from synthetic datasets to real-world deployment, future research should proceed in three strategic directions:

1. Architectural Upgrades for Global Receptive Fields:

The simplistic CNN Summary Network is constrained by its localized convolutional kernels, which may struggle to integrate widely separated arc fragments of a single lensed source. Future iterations should upgrade to a higher-capacity architecture, such as a Vision Transformer (ViT) [7] or an attention-augmented ResNet. The self-attention mechanism in ViTs would allow the summary network to capture global morphological dependencies across the entire 64×64 grid simultaneously, thereby preserving multi-scale spatial features before feeding them into the cINN. This would dramatically improve the image reconstruction fidelity without sacrificing inference speed.

2. Expanding Physical Priors for Realistic Cosmologies:

Real-universe gravitational lenses are rarely isolated, perfectly symmetric, and unperturbed elliptical masses. The Singular Isothermal Ellipsoid (SIE) profile used in this study, while robust, is an oversimplification. Future simulators must expand the parameter space to include more complex and realistic physics. This includes modeling external shear (γ_1, γ_2) induced by neighboring galaxy clusters along the line of sight, and adopting the Navarro-Frenk-White (NFW) profile to model extended dark matter halos. Furthermore, incorporating dark matter substructures (subhalos) into the forward modeling would allow the cINN to detect minute flux anomalies, turning this pipeline into a powerful tool for probing the fundamental nature of dark matter.

3. Advanced Domain Adaptation for Observational Data:

Despite injecting extreme noise and PSF blurring, training exclusively on synthetic data inherently risks domain shift when applied to real telescope images (e.g., from the James Webb Space Telescope or the Euclid mission). Future work should integrate Domain Adversarial Neural Networks (DANN) or contrastive self-supervised learning techniques. By concurrently processing unlabeled real telescope images alongside labeled synthetic data, the summary network can be forced to learn domain-invariant feature representations, effectively neutralizing the Sim-to-Real gap and ensuring that the cINN's posterior calibration remains flawless on actual observational datasets.

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